Status	Finished
Started	Monday, 3 November 2025, 7:40 PM
Completed	Monday, 3 November 2025, 8:10 PM
Duration	30 mins

Question **1**

Correct

The k-digit number N is an Armstrong number if and only if the k-th power of each digit sums to N.

Given a positive integer N, return true if and only if it is an Armstrong number.

Example 1:

Input:

153

Output:

true

Explanation:

153 is a 3-digit number, and 153 = 1^3 + 5^3 + 3^3.

Example 2:

Input:

123

Output:

false

Explanation:

123 is a 3-digit number, and 123 != 1^3 + 2^3 + 3^3 = 36.

Example 3:

Input:

1634

Output:

true

Note:

1 <= N <= 10^8

Answer: (penalty regime: 0 %)

```
#include<stdio.h>
#include<math.h>
int main()
{
int n,i,nc,nc1,ct=0;
int rev=0;
scanf("%d",&n);
nc=n;
while(nc>0)
{
nc=nc/10;
ct++;
}
nc1=n;
while(nc1>0)
{
i=nc1%10;
rev=rev+(int)pow(i,ct);
nc1=nc1/10;
}
if(rev==n)
{
printf("true");
}
else
{
printf("false");
}
return 0;
}

```

	Input	Expected	Got	
✓	153	true	true	✓
✓	123	false	false	✓

Passed all tests! ✓

Question **2**

Correct

Take a number, reverse it and add it to the original number until the obtained number is a palindrome.

Constraints

1<=num<=99999999

Sample Input 1

32

Sample Output 1

55

For example:

Input	Result
32	55
1234	5555

Answer: (penalty regime: 0 %)

```
#include<stdio.h>
int main()
{
    int n,m;
    int o=0;
    int p=0;
    scanf("%d",&n);
    do
    {
        o=n;
        m=0;
        while(n!=0)
        {
            m=(m*10)+(n%10);
            n=n/10;
        }

        n=o+m;
        p++;
    }
    while(m!=o||p==1);
    {
        printf("%d",m);
    }
```

```
    return 0;
}
```

	Input	Expected	Got	
✓	32	55	55	✓
✓	1234	5555	5555	✓

Passed all tests! ✓

Question **3**

Correct

Maya, a student in an arts and crafts class, wants to create a pattern using stars (*) in a specific format. She plans to use a program to help her construct the pattern.
 Write a program that takes an integer as input and constructs the following pattern using nested for loops.

Input: 5

Output:

```
*
* *
* * *
* * * *
* * * * *
* * * *
* * *
* *
*
```

Answer: (penalty regime: 0 %)

```
#include<stdio.h>
int main()
{
    int n,i,j;
    scanf("%d",&n);
    for (i=1;i<=n;i++)
    {
        for(j=1;j<=i;j++)
        {
            printf("* ");
        }
        printf("\n");
    }
    for (i=n-1;i>=1;i--)
    {
        for(j=1;j<=i;j++)
        {
            printf("* ");
        }
    printf("\n");
}
return 0;
}
```

	Input	Expected	Got	
✓	5	* *	* *	✓

Passed all tests! ✓